

Did Scientists Just Spot Dark Matter? One Clue Has Physicists Excited

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Dark matter is an invisible substance that scientists think makes up most of the matter in the universe. They can sense its **gravity** tugging on stars and other objects in space. But no one knows what it is, and no one has ever directly observed it, despite decades of searching.

Now, a detector called LUX-ZEPLIN, or LZ, may have glimpsed it for the first time. But scientists say they cannot be sure yet.

LZ is located at the Sanford Underground Research Facility in Lead, South Dakota. It searches for proposed **particles** called WIMPs, short for weakly interacting massive **particles**. Some scientists think WIMPs might make up dark matter. If a WIMP crashed into the center, or **nucleus**, of an atom in LZ's tank of liquid **xenon**, it would send that **nucleus** zinging away and give off a tiny flash of light and **electrons**. Sensors in the detector are designed to spot those signals.

The team combed through 220 days' worth of data, looking for a type of WIMP that would push **xenon** nuclei away at high energies. They found just one event that fits. It was recorded in June 2023. The team shared its findings on September 1 at a scientific meeting in Japan.

"We got left with one event, which is fascinating. Absolutely fascinating," said Rick Gaitskell, a **physicist** at Brown University and a spokesperson for LZ.

The team estimates there is only about a 1 in 100 chance that known **particles** caused the event. **Physicists** usually need much stronger proof before they say they have found evidence of something new, so more data is needed.

Wick Haxton, a **physicist** at the University of California, Berkeley, who was not part of the research, called it "the most interesting thing that's come up in recent times."

Word Treasure

gravity -- The force that pulls objects toward each other.

particles -- Extremely tiny pieces of matter.

nucleus -- The tiny, heavy center of an atom.

xenon -- A rare chemical element that is usually a gas but can be cooled into a liquid.

electrons -- Tiny particles with a negative electric charge that move around an atom's nucleus.

physicist -- A scientist who studies matter, energy and how forces work.

Background you need

Dark matter is the name scientists give to unseen material whose gravity seems to help hold galaxies together and affect how stars move.

Because it does not give off or reflect light, scientists try to catch it in other ways, such as watching for rare bumps between dark matter particles and ordinary atoms.

Detectors for these searches are often built deep underground, where layers of rock block many other particles that could cause false signals.

WIMPs are one popular idea for what dark matter particles might be.

Break it down

LEAD: The LZ detector may have glimpsed dark matter for the first time, but scientists cannot be sure

BACKGROUND: Dark matter's gravity can be sensed, but it has never been directly observed

KEY EVENT: One possible WIMP event found in 220 days of data

+ - WHERE: Sanford Underground Research Facility in Lead, South Dakota

+ - WHEN: Recorded in June 2023; shared September 1 at a meeting in Japan

EVIDENCE: About a 1 in 100 chance that known particles caused the event

OPEN QUESTION: More data is needed before scientists can claim a discovery

Why it matters

Finding dark matter would help explain what most of the matter in the universe is made of. The story also shows how scientists wait for strong proof before announcing a discovery.

Test what you remember

1. What is dark matter, according to the article?

- ☐ (A) A kind of black rock on the Moon
- ☐ (B) A new type of telescope
- ☐ (C) An invisible substance thought to make up most matter in the universe
- ☐ (D) A cloud of dust near Earth

2. Where is the LZ detector located?

- ☐ (A) Providence, Rhode Island
- ☐ (B) Lead, South Dakota
- ☐ (C) Berkeley, California
- ☐ (D) Tendo, Japan

3. What does WIMP stand for?

- ☐ (A) Weakly interacting massive particle
- ☐ (B) Warm invisible mass point
- ☐ (C) Wide imaging matter probe
- ☐ (D) Water ice moving particle

4. How many days of data did the team search?

- ☐ (A) 100
- ☐ (B) 2,023
- ☐ (C) 220
- ☐ (D) 23

5. How many events matching the search did the team find?

- ☐ (A) One
- ☐ (B) None
- ☐ (C) Two
- ☐ (D) Ten

6. What chance did the team estimate that known particles caused the event?

- ☐ (A) About 1 in 2
- ☐ (B) About 1 in 100
- ☐ (C) About 1 in 10
- ☐ (D) No chance at all

Answer key (try the quiz first!): 1: C 2: B 3: A 4: C 5: A 6: B

Questions to think about

1. Rick Gaitskell, LZ spokesperson: Calls the single remaining event absolutely fascinating.
2. Wick Haxton, physicist not involved in the study: Calls it the most interesting thing to come up in recent times.

MY THOUGHTS (at least 20 words):